An Analysis of Obesity Prevalence and Associated Factors in Scotland (2008–2012)

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1 Introduction

Obesity is a major public health concern, associated with an increased risk of chronic conditions such as cardiovascular disease, type 2 diabetes, and certain cancers. Understanding how obesity prevalence changes over time, and how it is influenced by demographic, socio-economic, and lifestyle factors, is essential for informing effective health policy and interventions. This report analyses data from the Scottish Health Surveys conducted between 2008 and 2012. The dataset includes individual-level information on age, sex, educational attainment, dietary behaviours (fruit and vegetable consumption), and obesity status. These variables provide a valuable opportunity to explore both temporal trends and potential risk factors associated with obesity within the Scottish population. The primary aim of this analysis is to assess whether the prevalence of obesity in Scotland has changed over the period 2008–2012. In addition, the report investigates whether obesity varies across key demographic groups, socio-economic indicators, and lifestyle behaviours.

2 Exploratory Analysis

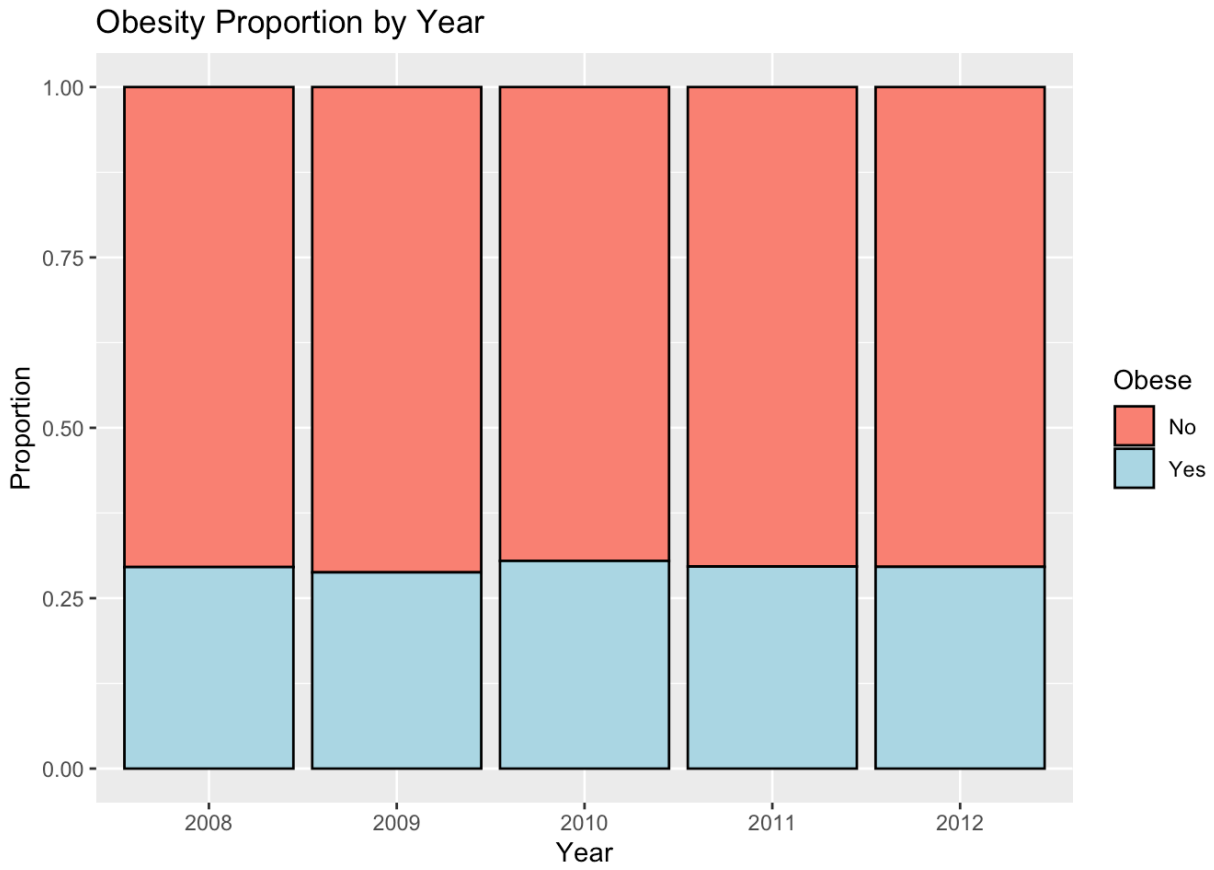


Figure 1: Obesity by Year Bar Plot

The bars in the plot appear to be fairly consistent. This indicates that the prevalence of obesity across Scotland over the years in the survey has not changed much, although it may be more clear from a table.

Table 1: prevelance comparison table

Obesity Prevalence by Year

Year	Obese	Total	Prevalence
2008	1448	4897	0.296
2009	1647	5721	0.288
2010	1666	5467	0.305
2011	1620	5464	0.296
2012	1088	3675	0.296

Now using a table we see that the prevalence of obesity is 0.296 in 2008, 2011 and 2012. It slightly dropped in 2009 and peaked in 2010 but not enough to infer that the prevalence changed over the years. Furthermore, these two outliers are still pretty close to 0.296.

Table 1: Obesity prevalence by sex, age group, education level, and lifestyle factors.

Subgroup	N	Obese (%)
Sex		
Female	13960	29.8
Male	11264	29.4
Age Group		
16–24	2063	15.0
25–34	3331	21.6
35–44	4485	29.9
45–54	4898	31.4
55–64	4566	36.0
65–74	3713	35.0
75+	2168	28.5
Education		
Degree or higher	6891	24.8
HNC/D or equiv	2598	28.8
Higher grade or equiv	3717	26.8
No qualifications	5298	36.5
Other school level	1875	33.4
Standard grade or equiv	4845	30.2
Lifestyle		
Veg: No, Fruit: No	2626	30.0
Veg: No, Fruit: Yes	2801	34.0
Veg: Yes, Fruit: No	5136	29.0
Veg: Yes, Fruit: Yes	14661	28.9

@summary-table breaks down obesity prevalence across the four variables of interest; the following plots examine each factor in turn.

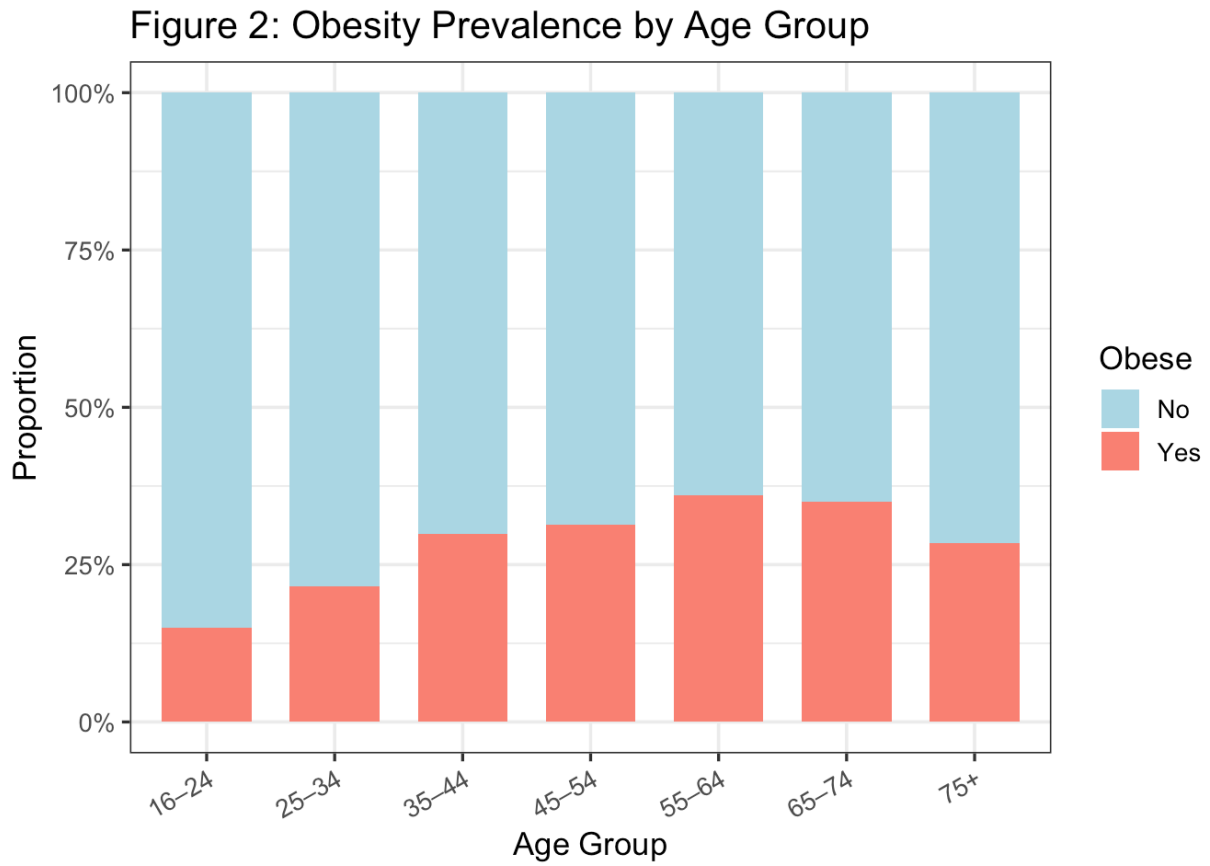


Figure 2

Figure 2 shows obesity rates climbing through middle age, reaching a peak around the 45–64 bracket, then dropping off for those aged 75 and over. The lower rate in the oldest group is worth treating with some caution, as it may partly reflect survivorship bias rather than a genuine health advantage.

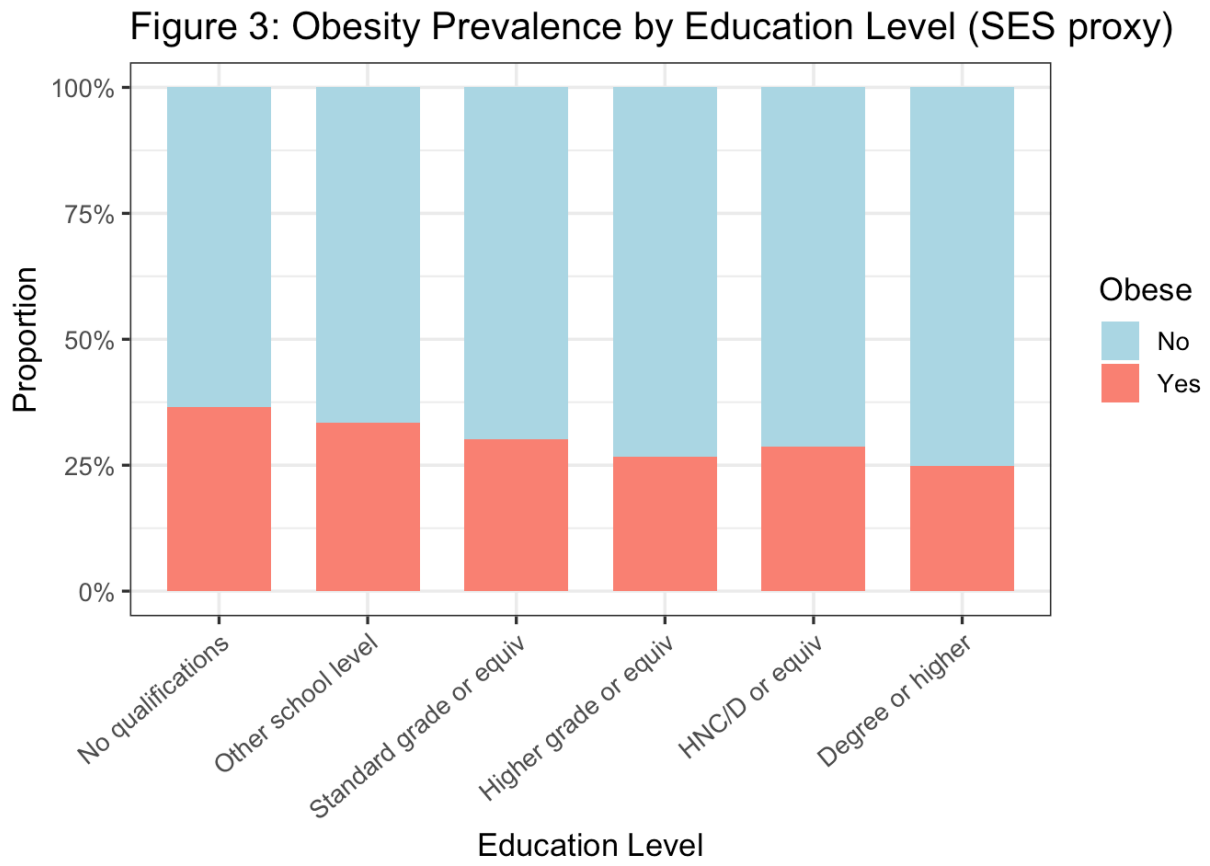


Figure 3

Figure 3 points to a fairly consistent socioeconomic gradient: those with no qualifications have the highest obesity rates, and the proportion falls as education level rises. This sits well with wider public health evidence linking lower socioeconomic position to worse health outcomes.

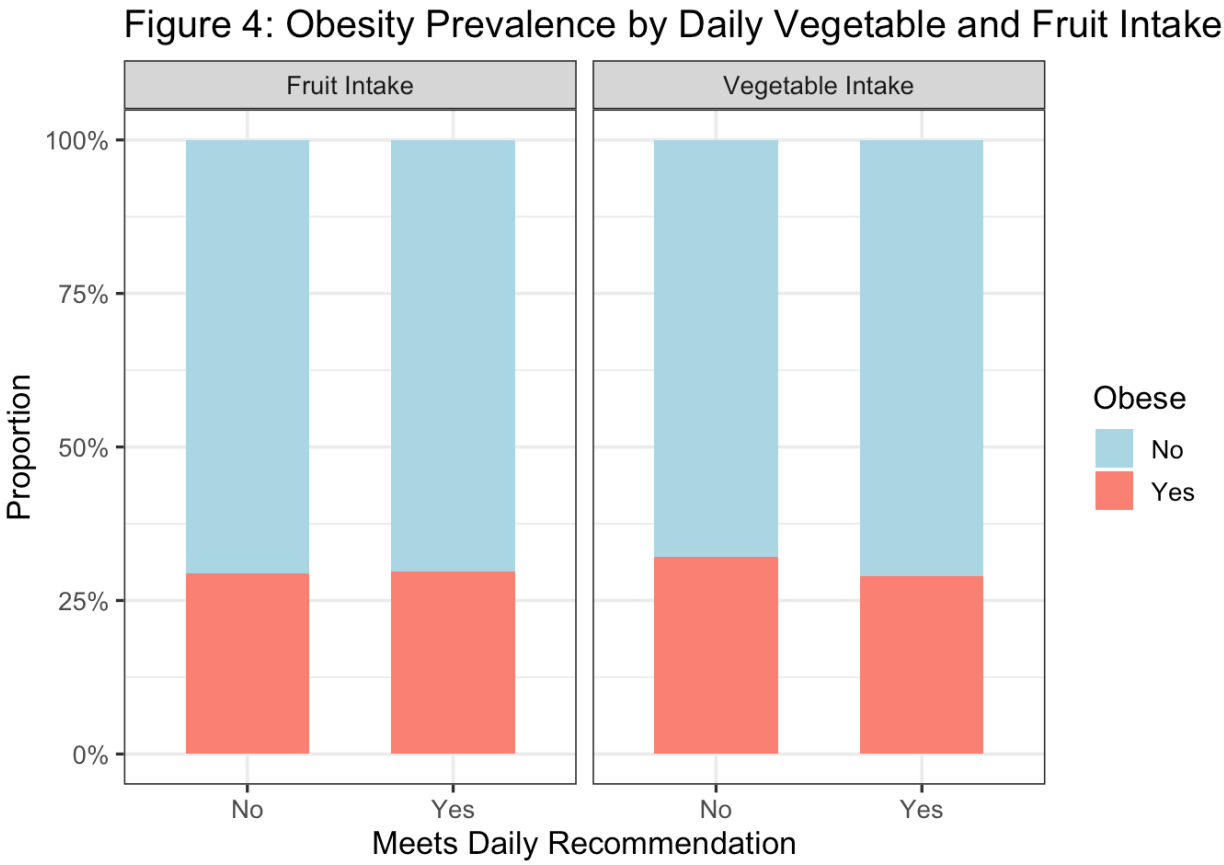


Figure 4

Figure 4 indicates that people who meet the daily recommended intake for both vegetables and fruit tend to have slightly lower obesity rates. The difference is not dramatic, but it holds for both dietary variables, hinting that behavioural factors may contribute alongside the demographic patterns seen earlier.

3 Formal Analysis

Since obesity status (Obese) is a binary outcome variable (Yes/No), a binary logistic regression model was used. First, survey year was included as a categorical predictor so that obesity prevalence could vary freely across the five survey years. The null hypothesis for this model was that obesity prevalence was the same across all years. A likelihood ratio test was used to assess the overall effect of year.

To assess whether there was evidence of a linear time trend, a second logistic regression model was fitted with year treated as a continuous numeric variable. Odds ratios (ORs), 95% confidence intervals, and p-values were used to interpret the results.

```
Call:
glm(formula = Obese ~ Year, family = binomial, data = dat)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.867901   0.031314 -27.716   <2e-16 ***
Year2009    -0.037769   0.042816  -0.882    0.378
Year2010     0.043062   0.042940   1.003    0.316
Year2011     0.003814   0.043104   0.088    0.929
Year2012     0.001743   0.047814   0.036    0.971
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 30649  on 25223  degrees of freedom
Residual deviance: 30645  on 25219  degrees of freedom
AIC: 30655

Number of Fisher Scoring iterations: 4
```

Logistic regression for obesity with year as a categorical predictor (ORs relative to 2008).

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.420	0.031	-27.716	0.000	0.395	0.446
Year2009	0.963	0.043	-0.882	0.378	0.885	1.047
Year2010	1.044	0.043	1.003	0.316	0.960	1.136
Year2011	1.004	0.043	0.088	0.929	0.923	1.092
Year2012	1.002	0.048	0.036	0.971	0.912	1.100

```
Analysis of Deviance Table

Model: binomial, link: logit

Response: Obese

Terms added sequentially (first to last)

     Df Deviance Resid. Df Resid. Dev Pr(>Chi)
NULL                 25223      30649
Year  4   3.8157     25219      30645   0.4315

Analysis of Deviance Table

Model 1: Obese ~ 1
Model 2: Obese ~ Year
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1     25223      30649
2     25219      30645  4   3.8157   0.4315


Call:
glm(formula = Obese ~ Year_num, family = binomial, data = dat)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -12.130043  20.709093  -0.586    0.558
Year_num      0.005604   0.010304   0.544    0.586

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 30649  on 25223  degrees of freedom
Residual deviance: 30648  on 25222  degrees of freedom
AIC: 30652

Number of Fisher Scoring iterations: 4
```

```
Call:
glm(formula = Obese ~ Year + Age + Sex + Education + Veg + Fruit,
    family = binomial, data = dat)

Coefficients:
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                     -1.1697450  0.0729605 -16.033  < 2e-16 ***
Year2009                        -0.0234869  0.0431662  -0.544  0.58637
Year2010                         0.0553755  0.0433262   1.278  0.20121
Year2011                         0.0219276  0.0435116   0.504  0.61430
Year2012                         0.0080077  0.0482806   0.166  0.86827
Age                              0.0116325  0.0009292  12.518  < 2e-16 ***
SexMale                         -0.0338321  0.0281358  -1.202  0.22919
EducationOther school level     -0.1678333  0.0570231  -2.943  0.00325 **
EducationStandard grade or equiv -0.0804781  0.0454689  -1.770  0.07673 .
EducationHigher grade or equiv  -0.2375041  0.0497808  -4.771 1.83e-06 ***
EducationHNC/D or equiv         -0.1447182  0.0545974  -2.651  0.00803 **
EducationDegree or higher       -0.3867856  0.0425161  -9.097  < 2e-16 ***
VegYes                          -0.1412367  0.0343153  -4.116 3.86e-05 ***
FruitYes                         0.0036470  0.0316223   0.115  0.90818
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 30649  on 25223  degrees of freedom
Residual deviance: 30247  on 25210  degrees of freedom
AIC: 30275

Number of Fisher Scoring iterations: 4

Warning in anova.glm(mod_year_adj, type = "II", test.statistic = "LR"): the
following arguments to 'anova.glm' are invalid and dropped: list(type = "II",
test.statistic = "LR")

Analysis of Deviance Table

Model: binomial, link: logit

Response: Obese

Terms added sequentially (first to last)

          Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                      25223      30649
Year       4    3.816     25219      30645    0.4315
Age        1  262.277     25218      30383 < 2.2e-16 ***
Sex        1    1.101     25217      30382    0.2940
Education  5  116.840     25212      30265 < 2.2e-16 ***
Veg        1   17.112     25211      30248 3.524e-05 ***
Fruit      1    0.013     25210      30248    0.9082
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Logistic regression testing linear yearly trend in obesity prevalence.

term	estimate	std.error	statistic	p.value	conf.low	conf.high

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.000	20.709	-0.586	0.558	0.000	2.300971e+12
Year_num	1.006	0.010	0.544	0.586	0.986	1.026000e+00

Adjusted logistic regression model for obesity.

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.310	0.073	-16.033	0.000	0.269	0.358
Year2009	0.977	0.043	-0.544	0.586	0.898	1.063
Year2010	1.057	0.043	1.278	0.201	0.971	1.151
Year2011	1.022	0.044	0.504	0.614	0.939	1.113
Year2012	1.008	0.048	0.166	0.868	0.917	1.108
Age	1.012	0.001	12.518	0.000	1.010	1.014
SexMale	0.967	0.028	-1.202	0.229	0.915	1.022
EducationOther school level	0.845	0.057	-2.943	0.003	0.756	0.945
EducationStandard grade or equiv	0.923	0.045	-1.770	0.077	0.844	1.009
EducationHigher grade or equiv	0.789	0.050	-4.771	0.000	0.715	0.869
EducationHNC/D or equiv	0.865	0.055	-2.651	0.008	0.777	0.963
EducationDegree or higher	0.679	0.043	-9.097	0.000	0.625	0.738
VegYes	0.868	0.034	-4.116	0.000	0.812	0.929
FruitYes	1.004	0.032	0.115	0.908	0.943	1.068

In the logistic regression model with year treated as a categorical variable, there was no evidence that obesity prevalence differed across survey years overall (likelihood ratio test: $\chi 2$=3.82, df=4, p=0.432).

Using 2008 as the reference year, none of the other survey years showed a statistically significant difference in the odds of obesity:

· 2009: p=0.378

· 2010: p=0.316

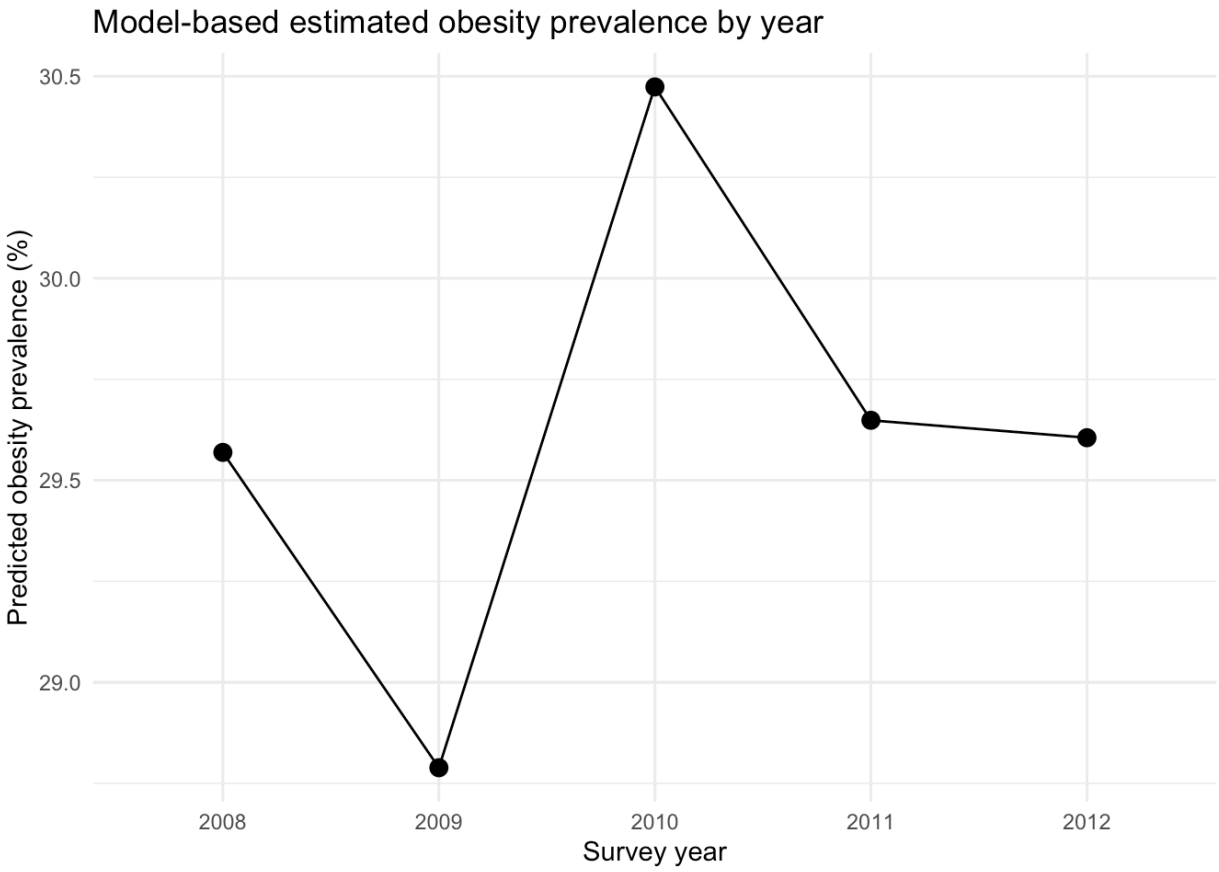
· 2011: p=0.929

· 2012: p=0.971

A second model was fitted to assess a linear trend over time. This also showed no evidence of a significant linear trend in obesity prevalence between 2008 and 2012 (p=0.586). The estimated odds ratio per additional year was approximately e0.005604≈1.006, suggesting almost no change in the odds of obesity from one year to the next.

Predicted probability of obesity by year from the unadjusted model.

Year	pred
2008	0.296
2009	0.288
2010	0.305
2011	0.296
2012	0.296



There is no statistically significant evidence that obesity prevalence in Scotland changed over the period 2008–2012. Both the categorical year model and the linear trend model suggest that obesity prevalence remained broadly stable across these survey years.

To investigate whether obesity differed by age, gender, socio-economic status and lifestyle factors, we fitted a binomial logistic regression model with obesity status as the response variable. The response variable Obese was binary, with Yes indicating an obese individual and No otherwise. The explanatory variables included age, sex, education level, vegetable intake and fruit intake. Logistic regression was appropriate because it models a binary outcome using the logit link, ensuring fitted probabilities remain within the unit interval.

There may be an interactive relationship between age and gender parameters in the model. Compare pure additive models and interaction models with age and gender using AIC. The AIC of the interaction model is smaller, so the interaction model should be chosen.

```
# Comparison of Model Performance Indices

Name        | Model |   AIC (weights)
-------------------------------------
model_add   |   glm | 30271.3 (0.004)
model_inter |   glm | 30260.2 (0.996)
```

$$Y_i \sim \text{Bernoulli}(p_i)$$

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 Age_i + \beta_2 Sex_i + \beta_3(Age_i \times Sex_i) + \sum_{j=1}^{5} \beta_{4j} Education_{ij} + \beta_5 Veg_i + \beta_6 Fruit_i$$

- Y_i is the obesity status of the ith individual (1 = obese, 0 = not obese);
- $p_i = P(Y_i = 1)$ is the probability that the ith individual is obese;

• β_0 is the intercept of the logistic regression model;

• β_1 represents the change in the log-odds of obesity for a one increase in age for female.

• β_2 represents the difference in the log-odds of obesity between the two sex groups when age is 0;

• β_3 represents the interaction effect between age and sex, that is, how the effect of age on the log-odds of obesity differs by sex;

• $Education_i$is a categorical variable describing the education level of the ith individual. The education categories included in the model are: Higher grade or equivalent, HNC/D or equivalent, No qualifications, Other school level, Standard grade or equivalent.

The reference category is Degree or higher.

• β_{4j}represents the change in the log-odds of obesity for each education category compared with the reference category (Degree or higher).

• β_5 and β_6 are the regression coefficients for vegetable intake and fruit intake respectively;

• $Age_i, Sex_i, Education_i, Veg_i, Fruit_i$are the explanatory variables for the ith individual.

	Obese		
Predictors	*Log-Odds*	*CI*	*p*
(Intercept)	0.36	0.00 – Inf	**<0.001**
Age	1.01	0.00 – Inf	**<0.001**
Sex [Male]	0.71	0.00 – Inf	**<0.001**
Education [Other school level]	0.85	0.00 – Inf	**0.004**
Education [Standard grade or equiv]	0.92	0.00 – Inf	0.069
Education [Higher grade or equiv]	0.79	0.00 – Inf	**<0.001**
Education [HNC/D or equiv]	0.86	0.00 – Inf	**0.006**
Education [Degree or higher]	0.68	0.00 – Inf	**<0.001**
Veg [Yes]	0.87	0.00 – Inf	**<0.001**
Fruit [Yes]	1.00	0.00 – Inf	0.948
Age × Sex [Male]	1.01	0.00 – Inf	**<0.001**
Observations	25224		
R^2 Tjur	0.016		

```
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
```

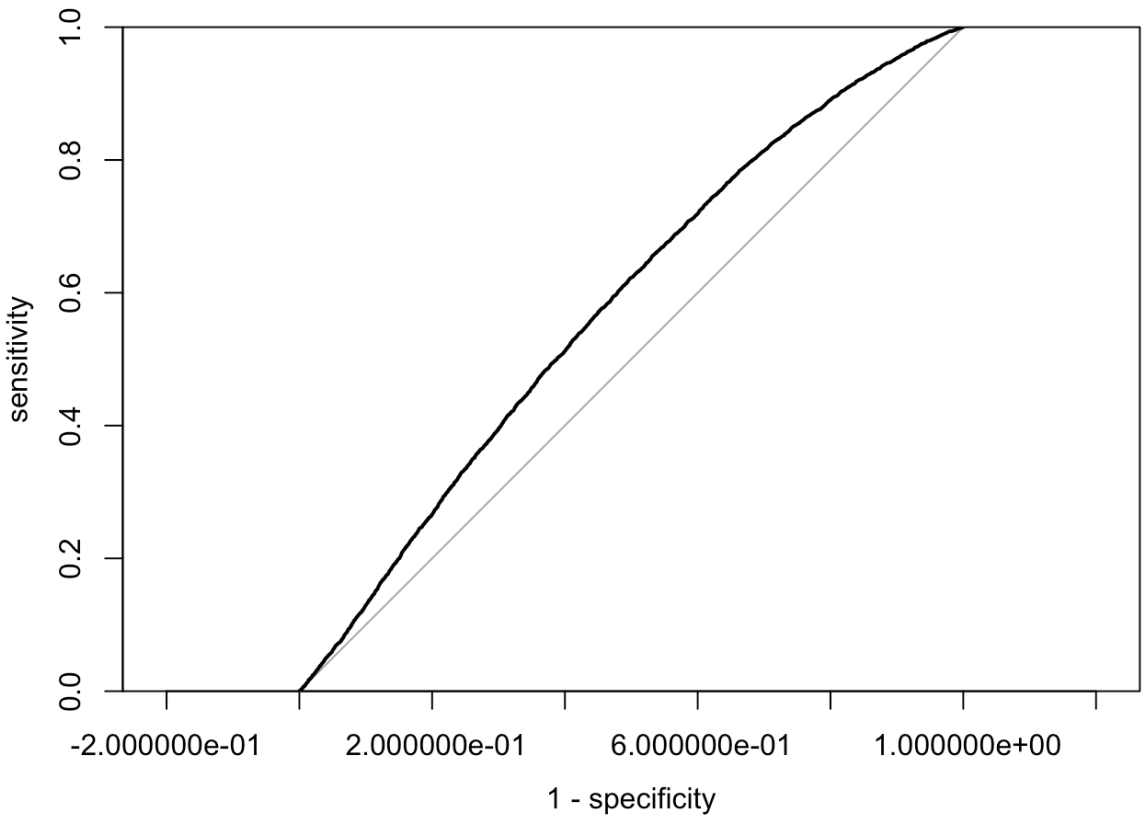
	Obese

Predictors	*Log-Odds*	*CI*	*p*
(Intercept)	-1.02	-1.17 – -0.87	**<0.001**
Age	0.01	0.01 – 0.01	**<0.001**
Sex [Male]	-0.34	-0.51 – -0.16	**<0.001**
Education [Other school level]	-0.16	-0.28 – -0.05	**0.004**
Education [Standard grade or equiv]	-0.08	-0.17 – 0.01	0.069
Education [Higher grade or equiv]	-0.24	-0.34 – -0.14	**<0.001**
Education [HNC/D or equiv]	-0.15	-0.26 – -0.04	**0.006**
Education [Degree or higher]	-0.39	-0.47 – -0.31	**<0.001**
Veg [Yes]	-0.14	-0.21 – -0.08	**<0.001**
Fruit [Yes]	0.00	-0.06 – 0.06	0.948
Age × Sex [Male]	0.01	0.00 – 0.01	**<0.001**
Observations	25224		
R^2 Tjur	0.016		

Since the p-value for fruit consumption is greater than 0.05 and the 95% confidence interval includes 0, fruit intake is not significantly associated with obesity. Apart from fruit, all other predictors are statistically significant(The 0.00 in the CI of the interaction item is due to rounding). This provides evidence that age, sex, education level, vegetable consumption and the interaction between age and sex are significantly associated with obesity in this model. Age has β = 0.01, 95% CI: 0.01–0.01, indicating that for females, each additional year of age increases the log-odds of being obese by approximately 0.01. The coefficient for Sex [Male] (β = −0.34, 95% CI: −0.51 to −0.16) represents the difference in log-odds of obesity between males and females when age is zero. The interaction between age and sex has β = 0.01, 95% CI: 0.00–0.01, indicating that the effect of age on obesity differs by sex and increases more strongly for males than for females. Vegetable has β = −0.14, 95% CI: −0.21 to −0.08, indicating that individuals who eat vegetables have lower log-odds of being obese compared with those who do not. All estimated coefficients for the education categories are positive relative to the reference category (Degree or higher), indicating that individuals in these groups have higher log-odds of obesity than those with a degree or higher qualification.

```
  threshold sensitivity specificity
1 0.2858467   0.6219039   0.5018305

# A tibble: 5 × 3
  .metric  .estimator .estimate
  <chr>    <chr>          <dbl>
1 accuracy binary         0.537
2 sens     binary         0.622
3 spec     binary         0.502
4 ppv      binary         0.344
5 npv      binary         0.759
```



The model-predicted values closely match the observed data. Most residuals fall within the expected error bounds, indicating no strong evidence of systematic lack of fit. No observations fall outside the contour lines, suggesting that there are no highly influential data points affecting the model estimates. Most predictors have relatively low VIF values, indicating that multicollinearity among the explanatory variables is not severe. The quantile residuals lie close to the reference line, suggesting that the residual distribution follows the expected pattern.

The best cut-off probability was approximately 0.28, which provides a balance between sensitivity and specificity. The model achieves an accuracy of 0.537, with sensitivity of 0.622 and specificity of 0.502, indicating moderate ability to identify obese individuals and have limited ability to correctly classify non-obese individuals. The ROC curve lies above the diagonal reference line, suggesting that the model performs better than random classification, although the overall discriminative ability remains relatively modest.

4 Conclusions

Obesity prevalence in Scotland is influenced by demographic and socio-economic factors, with higher rates observed among older individuals and those with lower educational attainment. While lifestyle factors such as diet show some association, their effects are less pronounced when other variables are considered. The logistic regression model captures these relationships to a limited extent, performing better at identifying obese individuals than non-obese ones. Overall, the findings suggest that obesity is shaped by multiple interacting factors and that effective interventions should focus on higher-risk groups and broader socio-economic conditions.